

IMO Training – Number Theory

Infinite Descent and Vieta Jumping

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Notes

Example. Find all positive integral solutions to the system $\begin{cases} a^2 + 6b^2 = p^2 \\ b^2 + 6a^2 = q^2. \end{cases}$

Example. Show that $\sqrt[3]{2}$ is irrational.

Example. Solve $x^3 + 2y^3 = 4z^3$ in non-negative integers.

Example. Prove that $x^4 + y^4 = z^4$ has no positive integral solution.

Corollary. For all positive integers k , prove that $x^{4k} + y^{4k} = z^{4k}$ has no positive integral solution.

Example. (Putnam 1988) Prove that $\forall n \in \mathbb{R}$, the equation

$$x_1^2 + x_2^2 + x_3^2 + x_4^2 = x_1x_2x_3 + x_1x_2x_4 + x_1x_3x_4 + x_2x_3x_4$$

has a solution for which x_1, x_2, x_3, x_4 are all integers larger than n .

Example. (IMO 2007) Let $a, b \in \mathbb{N}$. Show that if $4ab - 1 \mid (4a^2 - 1)^2$, then $a = b$.

Example. Let $x, y \in \mathbb{N}$ such that $xy \mid x^2 + y^2 + 1$. Prove that $\frac{x^2 + y^2 + 1}{xy} = 3$.

Example. Let $a, b \in \mathbb{N}$ with $ab \neq 1$. Suppose that $ab - 1 \mid a^2 + b^2$. Show that $\frac{a^2 + b^2}{ab - 1} = 5$.

Example. (Crux) If $a, b, c \in \mathbb{N}$ such that $0 < a^2 + b^2 - abc \leq c$, show that $a^2 + b^2 - abc$ is a perfect square.

Example. (IMO 1988 Problem 6) Let $a, b \in \mathbb{N}$ such that $ab + 1 \mid a^2 + b^2$. Prove that $\frac{a^2 + b^2}{ab + 1}$ is a perfect square.

Extra Space